In [1]:
```
import pandas as pd
```

In [5]:
```
import json

with open("steam_all_2005.json", "r", encoding="utf-8") as f:
    data = json.load(f)

print(type(data))
```

```
<class 'dict'>
```

In [6]:
```
if isinstance(data, dict):
    print("Number of keys:", len(data))
    first_key = list(data.keys())[0]
    print("First key:", first_key)
    print(data[first_key])

elif isinstance(data, list):
    print("Number of items:", len(data))
    print(data[0])
```

```
Number of keys: 5
First key: ratings
['Overwhelmingly Positive', 'Very Positive', 'Mostly
Positive', 'Positive', 'Mixed', 'Mostly Negative',
'Negative', 'Very Negative', 'Overwhelmingly Negative']
```

In [7]:
```
print(data.keys())
```

```
dict_keys(['ratings', 'genres', 'tags', 'games', 'meta'])
```

In [8]:
```
for key in data.keys():
    print(f"\n{key}")
    print(type(data[key]))

    if hasattr(data[key], "__len__"):
        print("Length:", len(data[key]))
```

```
ratings
<class 'list'>
Length: 9

genres
<class 'list'>
Length: 33

tags
<class 'list'>
Length: 50

games
<class 'list'>
Length: 82928

meta
<class 'dict'>
Length: 4
```

In [9]:
```
print(data["games"][0])
```

```
['Counter-Strike 2', 2012, 87, 8815087, 0.0, 1, [1, 8], [28,
18, 22, 2], 'Valve']
```

In [10]:

```
import pandas as pd

df = pd.DataFrame(data["games"])

df.head()
```

Out[10]:

	0	1	2	3	4	5	6	7
0	Counter-Strike 2	2012	87	8815087	0.00	1	[1, 8]	[28, 18, 22, 2]
1	PUBG: BATTLEGROUNDS	2017	59	2557944	0.00	4	[1, 3, 11, 8]	[46, 18, 22, 28]
2	Dota 2	2013	82	2498969	0.00	1	[1, 5, 8]	[24, 22, 4]
3	Grand Theft Auto V Legacy	2015	87	1990556	0.00	1	[1, 3]	[2, 22]
4	Terraria	2011	97	1409473	4.99	0	[1, 3, 0, 6]	[45, 46, 8, 22]

In [11]:

```
df.columns.tolist()
```

Out[11]:

```
[0, 1, 2, 3, 4, 5, 6, 7, 8]
```

In [12]:

```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 82928 entries, 0 to 82927
Data columns (total 9 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   0       82928 non-null  str
 1   1       82928 non-null  int64
 2   2       82928 non-null  int64
 3   3       82928 non-null  int64
 4   4       82928 non-null  float64
 5   5       82928 non-null  int64
 6   6       82928 non-null  object
 7   7       82928 non-null  object
 8   8       82928 non-null  str
dtypes: float64(1), int64(4), object(2), str(2)
memory usage: 8.3+ MB
```

In [15]:

```
from pprint import pprint

pprint(data["meta"])
```

```
{'count': 82928,
 'genre_matched': 82813,
 'source': 'fronkongames/steam-games-dataset (Jan 2026)',
 'yearRange': [2005, 2025]}
```

In [16]:

```
print(data["genres"])
```

```
['Indie', 'Action', 'Casual', 'Adventure', 'Simulation',
'Strategy', 'RPG', 'Early Access', 'Free To Play', 'Sports',
'Racing', 'Massively Multiplayer', 'Utilities', 'Design &
Illustration', 'Violent', 'Animation & Modeling',
'Education', 'Video Production', 'Gore', 'Game Development',
'Audio Production', 'Software Training', 'Photo Editing',
'Nudity', 'Web Publishing', 'Sexual Content', 'Accounting',
'Movie', 'Documentary', 'Episodic', 'Short', 'Tutorial', '360
Video']
```

In [17]:

```
print(data["tags"][:10])
```

```
['Indie', 'Casual', 'Action', 'Adventure', 'Strategy',
'Simulation', 'Puzzle', 'RPG', '2D', 'Singleplayer']
```

In [18]:

```
for i in range(5):
    print(data["games"][i])
```

```
['Counter-Strike 2', 2012, 87, 8815087, 0.0, 1, [1, 8], [28,
18, 22, 2], 'Valve']
['PUBG: BATTLEGROUNDS', 2017, 59, 2557944, 0.0, 4, [1, 3, 11,
8], [46, 18, 22, 28], 'PUBG Corporation']
['Dota 2', 2013, 82, 2498969, 0.0, 1, [1, 5, 8], [24, 22, 4],
'Valve']
['Grand Theft Auto V Legacy', 2015, 87, 1990556, 0.0, 1, [1,
3], [2, 22], 'Rockstar North']
['Terraria', 2011, 97, 1409473, 4.99, 0, [1, 3, 0, 6], [45,
46, 8, 22], 'Re-Logic']
```

In [21]:

```
df.columns = [
    "name",           # Game name
    "release_year",   # Release year
    "review_score",   # Review score (0-100)
    "review_count",   # Number of reviews
    "price",          # Price in USD
    "rating_id",      # Encoded review category (to be verified)
    "genre_ids",      # List of genre indices
    "tag_ids",        # List of tag indices
    "developer"       # Developer
]
```

In [22]:

```
df.head()
```

Out[22]:

	name	release_year	review_score	review_count
0	Counter-Strike 2	2012	87	8815087
1	PUBG: BATTLEGROUNDS	2017	59	2557944
2	Dota 2	2013	82	2498969
3	Grand Theft Auto V Legacy	2015	87	1990556
4	Terraria	2011	97	1409473

In [23]:

```
genres = data["genres"]

df["genres"] = df["genre_ids"].apply(
    lambda ids: [genres[i] for i in ids]
)
```

In [25]:

```
df[["genre_ids", "genres"]].head()
```

Out[25]:

	genre_ids	genres
0	[1, 8]	[Action, Free To Play]
1	[1, 3, 11, 8]	[Action, Adventure, Massively Multiplayer, Fre...
2	[1, 5, 8]	[Action, Strategy, Free To Play]
3	[1, 3]	[Action, Adventure]
4	[1, 3, 0, 6]	[Action, Adventure, Indie, RPG]

In [26]:

```
df.head()
```

Out[26]:

	name	release_year	review_score	review_count
0	Counter-Strike 2	2012	87	8815087
1	PUBG: BATTLEGROUNDS	2017	59	2557944
2	Dota 2	2013	82	2498969
3	Grand Theft Auto V Legacy	2015	87	1990556
4	Terraria	2011	97	1409473

In [27]:

```
tags = data["tags"]

df["tags"] = df["tag_ids"].apply(
    lambda ids: [tags[i] for i in ids]
)
```

In [28]:

```
df.head()
```

Out[28]:

	name	release_year	review_score	review_count
0	Counter-Strike 2	2012	87	8815087
1	PUBG: BATTLEGROUNDS	2017	59	2557944
2	Dota 2	2013	82	2498969
3	Grand Theft Auto V Legacy	2015	87	1990556
4	Terraria	2011	97	1409473

In [29]:

```
df["primary_genre"] = df["genres"].apply(
    lambda x: x[0] if len(x) > 0 else "Unknown"
)
```

In [30]:

```
df[["name", "primary_genre"]].head()
```

Out[30]:

	name	primary_genre
0	Counter-Strike 2	Action
1	PUBG: BATTLEGROUNDS	Action
2	Dota 2	Action
3	Grand Theft Auto V Legacy	Action
4	Terraria	Action

In [31]:

```
df.isnull().sum()
```

Out[31]:

```
name             0
release_year     0
review_score     0
review_count     0
price            0
rating_id        0
genre_ids        0
tag_ids          0
developer        0
genres           0
tags             0
primary_genre    0
dtype: int64
```

In [32]:

```
df.describe()
```

Out[32]:

	release_year	review_score	review_count	price
count	82928.000000	82928.000000	8.292800e+04	82928.000000
mean	2020.478319	75.829985	1.782995e+03	5.158642
std	3.302634	23.877829	3.964447e+04	11.701896
min	2005.000000	0.000000	1.000000e+00	0.000000
25%	2018.000000	65.000000	6.000000e+00	0.990000
50%	2021.000000	82.000000	2.400000e+01	2.990000
75%	2023.000000	94.000000	1.240000e+02	5.990000
max	2025.000000	100.000000	8.815087e+06	999.980000

In [33]:

```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 82928 entries, 0 to 82927
Data columns (total 12 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   name           82928 non-null  str
 1   release_year   82928 non-null  int64
 2   review_score   82928 non-null  int64
 3   review_count   82928 non-null  int64
 4   price          82928 non-null  float64
 5   rating_id      82928 non-null  int64
 6   genre_ids      82928 non-null  object
 7   tag_ids        82928 non-null  object
 8   developer      82928 non-null  str
 9   genres         82928 non-null  object
 10  tags           82928 non-null  object
 11  primary_genre  82928 non-null  str
dtypes: float64(1), int64(4), object(4), str(3)
memory usage: 10.7+ MB
```

In [35]:

```
print(df.shape)

print(df["primary_genre"].value_counts().head(10))

print(df["release_year"].describe())

print(df["price"].describe())

print(df["review_score"].describe())

print(df["developer"].nunique())
```

```
(82928, 12)
primary_genre
Action          35127
Adventure       17484
Casual          15006
Indie            7982
Simulation       1791
RPG              1326
Strategy         1225
Free To Play      509
Racing            440
Violent           340
Name: count, dtype: int64
count    82928.000000
mean      2020.478319
std          3.302634
min       2005.000000
25%       2018.000000
50%       2021.000000
75%       2023.000000
max       2025.000000
Name: release_year, dtype: float64
count    82928.000000
mean         5.158642
std         11.701896
min          0.000000
25%          0.990000
50%          2.990000
75%          5.990000
max        999.980000
Name: price, dtype: float64
count    82928.000000
mean        75.829985
std         23.877829
min          0.000000
25%         65.000000
50%         82.000000
75%         94.000000
max        100.000000
Name: review_score, dtype: float64
```

```
52472
```

In [36]:

```
df.to_pickle("steam_cleaned.pkl")

# Save as CSV
df.to_csv("steam_cleaned.csv", index=False)

print("Dataset saved successfully!")
```

```
Dataset saved successfully!
```

In [37]:

```
df.columns
```

Out[37]:

```
Index(['name', 'release_year', 'review_score',
'review_count', 'price',
       'rating_id', 'genre_ids', 'tag_ids', 'developer',
'genres', 'tags',
       'primary_genre'],
      dtype='str')
```

In [38]:

```
df.head()
```

Out[38]:

	name	release_year	review_score	review_count
0	Counter-Strike 2	2012	87	8815087
1	PUBG: BATTLEGROUNDS	2017	59	2557944
2	Dota 2	2013	82	2498969
3	Grand Theft Auto V Legacy	2015	87	1990556
4	Terraria	2011	97	1409473

In [ ]: